

Input
 $X: [B, D]$

RealNVP block stack
 K coupling steps

$z_K: [B, D]$

DiagGaussian
 $z_K \sim N(\mathbf{0}, \mathbf{I}_D)$

GaussianPCA
 $z_K \sim N(\mu, WW^T + \sigma^2 \mathbf{I}_D)$
 $\mu \sim N(\mathbf{0}, \mathbf{I}_M), \epsilon \sim N(\mathbf{0}, \mathbf{I}_D)$
 $z_K = \mu + W\mu + \sigma\epsilon$